

Abschnitt A

```
const int volumePin =  
A0;  
int volume;
```

```
const int strobePin = 7;  
  
const int resetPin = 8;  
const int outPin = A5;  
int level[7];
```

```
const int redPin1 = 3;  
const int greenPin1 = 5;  
const int bluePin1 = 6;
```

```
const int redPin2 = 9;  
const int greenPin2 = 10;  
const int bluePin2 = 11;
```

Abschnitt B

void setup() {		
	Serial.begin(9600);	
	pinMode(strobePin, OUTPUT); pinMode(resetPin, OUTPUT); pinMode(outPin, INPUT);	
	pinMode(redPin1, OUTPUT); pinMode(greenPin1, OUTPUT); pinMode(bluePin1, OUTPUT);	
	pinMode(redPin2, OUTPUT); pinMode(greenPin2, OUTPUT); pinMode(bluePin2, OUTPUT);	
	digitalWrite(resetPin, LOW); digitalWrite(strobePin, LOW); delay(1);	
	digitalWrite(resetPin, HIGH); delay(1); digitalWrite(resetPin, LOW); digitalWrite(strobePin, HIGH); delay(1);	

Abschnitt C

void loop() {		
volume = abs(analogRead(volumePin) - 512);		
	if (volume < 165) { analogWrite(redPin1, map(volume, 0, 164, 0, 255)); analogWrite(greenPin1, 0); analogWrite(bluePin1, 0); } else if (volume < 341) { analogWrite(redPin1, 0); analogWrite(greenPin1, map(volume, 171, 340, 0, 255)); analogWrite(bluePin1, 0); } else { analogWrite(redPin1, 0); analogWrite(greenPin1, 0); analogWrite(bluePin1, map(volume, 341, 512, 0, 255)); } 	
	digitalWrite(resetPin, HIGH); digitalWrite(resetPin, LOW);	
	for (int i = 0; i < 7; i++) { digitalWrite(strobePin, LOW); delayMicroseconds(30); level[i] = analogRead(outPin); digitalWrite(strobePin, HIGH); delayMicroseconds(30); } 	
	for (int i = 0; i < 7; i++) { Serial.print(level[i]); Serial.print(" "); } Serial.println();	
	int bass = level[0]; int mid = level[3]; int hoch = level[6];	
	analogWrite(redPin2, map(bass, 5, 25, 0, 255)); analogWrite(greenPin2, map(mid, 200, 1023, 0, 255)); analogWrite(bluePin2, map(hoch, 400, 700, 0, 255));	
	delay(10); }	

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Wiederholung